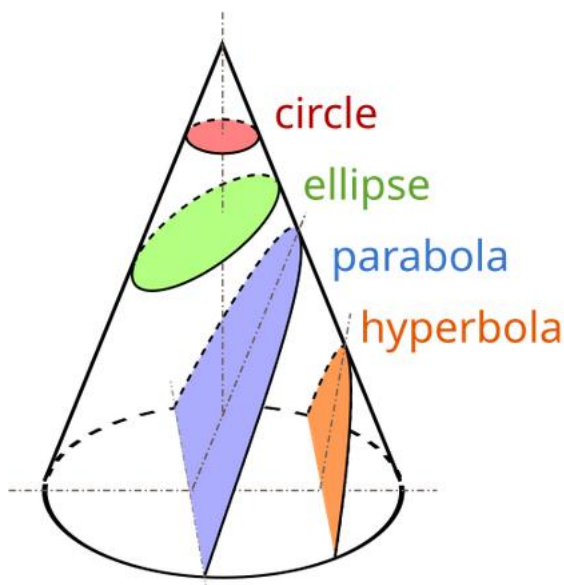


General Equation of Second Degree

Conics

A conic section, conic, or a quadratic curve is a curve obtained from a cone's surface intersecting a plane. The three types of conic sections are the hyperbola, the parabola, and the ellipse; the circle is a special case of the ellipse, though it was sometimes considered a fourth type.



Identification:

The general equation of the 2nd degree is,

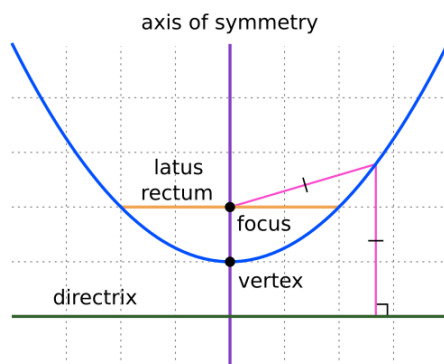
$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \dots\dots\dots (1)$$

If $\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$ then (1) represents

- (a) Circle if $\Delta \neq 0$ and $a = b, h = 0$
- (b) Parabola if $\Delta \neq 0$ and $ab - h^2 = 0$
- (c) Ellipse if $\Delta \neq 0$ and $ab - h^2 > 0$
- (d) Hyperbola if $\Delta \neq 0$ and $ab - h^2 < 0$

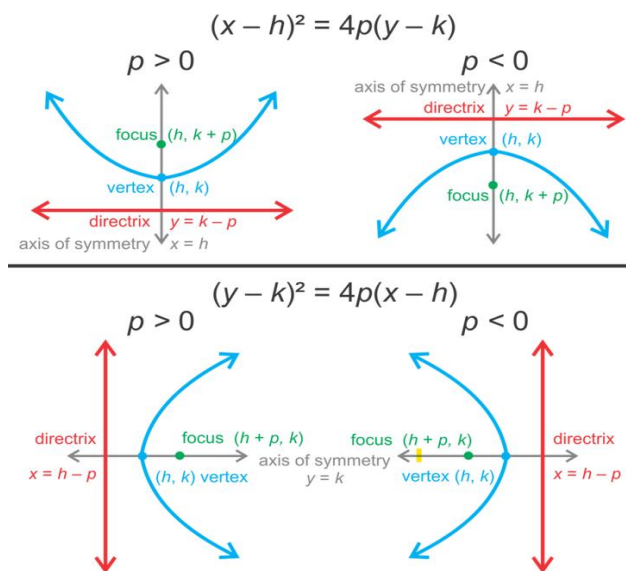
Parabola:

A parabola is a curve in which every point on the curve is equidistant from a point called a focus and a straight line called a directrix. It is a U-shaped curve that appears often in mathematics.



The general equation of a parabola is often expressed as $(x - h)^2 = 4p(y - k)$ or $(y - k)^2 = 4p(x - h)$

Where (h, k) is the vertex.



The standard equation of a regular parabola is $y^2 = 4ax$.

Equation of a tangent to the Parabola

The equation of the tangent to a parabola depends on the form of the parabola. For the standard parabola $y^2 = 4ax$, the tangent equation can be written in slope form as $y = mx + \frac{a}{m}$ and

in point form as $yy_1 = 2a(x + x_1)$ at (x_1, y_1) .

Ex 9. Reduce the equation $4x^2 - 4xy + y^2 - 8x - y + 6 = 0$ to the standard form. Find all its properties.

In the eq. (1), $a = 4, b = 1, h = -2, g = -4, f = -3, c = 5$

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = -64, ab - h^2 = 4 - 4 = 0$

$\Delta \neq 0$ and $ab = h^2$. Hence eq. (1) represents a parabola. The eq. (1) may be written as

$(2x - y)^2 = 8x + 6y - 5$... (2)

or, $(2x - y + \lambda)^2 = 8x + 6y - 5 + \lambda^2 + 2\lambda(2x - y)$... (3)

or, $(2x - y + \lambda)^2 = x(4\lambda + 8) + y(-2\lambda + 6) + (\lambda^2 - 5)$... (4)

The lines $2x - y + \lambda = 0$... (5)

and $x(4\lambda + 8) + y(-2\lambda + 6) + (\lambda^2 - 5) = 0$...

may be perpendicular if $a_1a_2 + b_1b_2 = 0$

i. e. $2(4\lambda + 8)(-1) + (-2\lambda + 6) = 0$ or, $\lambda = -1$... (6)

The eq. (3) becomes

$(2x - y - 1)^2 = 4(x + 2y - 1)$ which can be written as

$\left(\frac{2x - y - 1}{\sqrt{2^2 + 1^2}}\right)^2 = 4\sqrt{5}\left(\frac{x + 2y - 1}{\sqrt{1^2 + 2^2}}\right)$

It is of the form $PN^2 = 4(\sqrt{5})/5$. PM.

or, $Y^2 = 4/\sqrt{5}$. X. which is the standard equation of a parabola.

(i) Latus rectum $= 4p = 4/\sqrt{5}$ or, $p = 1/\sqrt{5}$

$X = (x + 2y - 1)/\sqrt{5}$ and $Y = (2x - y - 1)/\sqrt{5}$

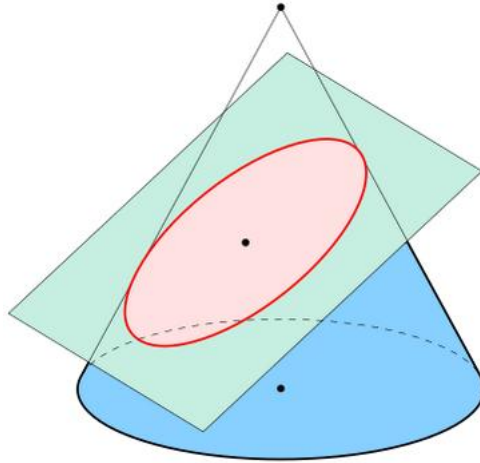
$\therefore Y^2 = 4p X$.

Exercise:

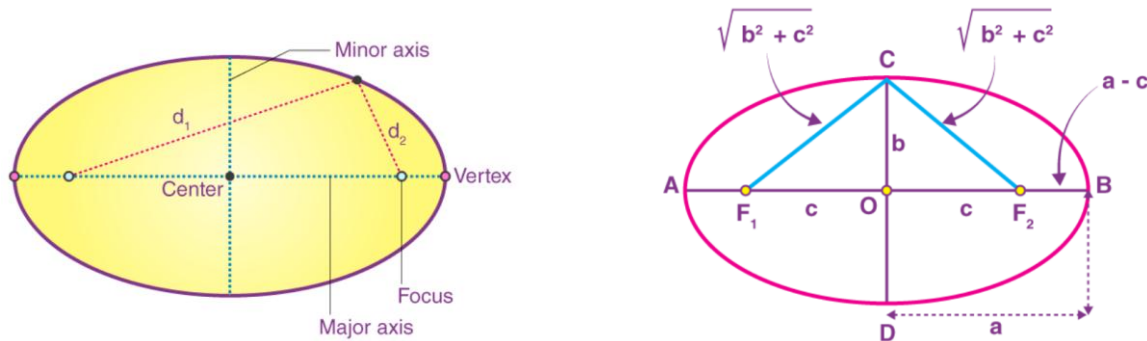
- Find the nature of the conic $x^2 + 2xy + y^2 - 6x - 2y + 4 = 0$. Reduce it to standard form. Ans: $\left(\frac{x+y-2}{\sqrt{2}}\right)^2 = 4 \cdot \frac{\sqrt{2}}{4} \left(\frac{x-y}{\sqrt{2}}\right)$

Ellipse

An ellipse is a closed, oval-shaped curve where the sum of the distances from any point on the curve to two fixed points (foci) is constant.



The ellipse is one of the conic sections, that is produced, when a plane cuts the cone at an angle with the base. If the cone is intersected by the plane, parallel to the base, then it forms a circle.



When the center of the ellipse is at the origin (0,0) and the foci are on the x-axis and y-axis, then

The equation of the ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Reduce to standard form (Ellipse)

Now the constant $= \frac{\Delta}{ab-h^2} = gx + fy + c$ which is represented by another constant c_1 .

$$c_1 = gx + fy + c$$

where x and y are found from the equations $\frac{\delta F}{\delta x} = 0$, and $\frac{\delta F}{\delta y} = 0$... (4)

and g, f and c are found from the given equation (1)

Hence the equation reduces to $ax^2 + 2hxy + by^2 + c_1 = 0$

In other words "Same terms of second degree as in the given equation + new constant = 0.
which we can always reduce to the FORM

$$Ax^2 + 2Hxy + By^2 = 1$$

... (5)

If the axes of the equation (5) are rotated about its new (centre) origin in such a way that xy -term vanishes, the new equation of the conic becomes $a_1x^2 + b_1y^2 + c_1 = 0$... (6)

which is called the Standard Form of the equation of the conic. Then by Invariants, (See Art. 35) we have from (5) and (6)

$$a_1 + b_1 = a + b$$

$$a_1b_1 = ab - h^2$$

where a_1 and b_1 can be easily found.

Thus the equation (1) reduces to the standard form

$$a_1x^2 + b_1y^2 + c_1 = 0$$

Ex. 14(a) What conic does the equation $5x^2 - 2xy + 5y^2 - 8x - 8y - 8 = 0$ represent? Reduce the equation to standard form. [N.U.H. 1996]

Ans. The given equation

$$5x^2 - 2xy + 5y^2 - 8x - 8y - 8 = 0 \quad \dots \quad (i)$$

Comparing (i) with $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

We get, $a = 5, h = -1, b = 5, g = -4, f = -4, c = -8$.

$$\therefore \Delta = abc + 2fgh - af^2 - bg^2 - ch^2$$

$$= 5 \cdot 5 \cdot (-8) + 2(-4)(-4)(-1) - 5(-4)^2 - 5(-4)^2 - (-8)(-1)^2$$

$$= -200 - 32 - 80 - 80 + 8 = -384$$

$$\therefore \Delta \neq 0$$

$$ab - h^2 = 5 \cdot 5 - (-1)^2 = 25 - 1 = 24$$

$$\therefore ab - h^2 > 0$$

Thus, the given conic represents an ellipse,

Let (α, β) be the centre of the conic (i)

$$\therefore \alpha = \frac{hf - bg}{ab - h^2}, \quad \beta = \frac{hg - af}{ab - h^2}$$

$$= \frac{(-1)(-4) - 5(-4)}{5.5 - (-1)^2}$$

$$= \frac{24}{24}$$

$$= 1$$

$$= \frac{(-1)(-4) - 5(-4)}{5.5 - (-1)^2}$$

$$= \frac{24}{24}$$

$$= 1$$

∴ Centre is at (1, 1)

Transfer the origin to a point (1,1) When direction of axes are unchanged, then (1) becomes

$$ax^2 + 2hxy + by^2 = c_1$$

$$\text{ie } 5x^2 - 2xy + 5y^2 = c_1$$

$$\text{where } c_1 = -(\alpha g + \beta f + c)$$

$$c_1 = -\{1 \cdot (-4) + 1 \cdot (-4) - 8\} = 16$$

$$\therefore 5x^2 - 2hxy + 5y^2 = 16$$

$$\Rightarrow \frac{5}{16}x^2 - 2 \cdot \frac{1}{16}xy + \frac{5}{16}y^2 = 1$$

$$\Rightarrow Ax^2 + 2Hxy + By^2 = 1 \dots \dots (2)$$

$$\text{Where } A = \frac{5}{16}, H = \frac{-1}{16}, B = \frac{5}{16}$$

Removing xy term when the axes being rotated.

$$\therefore a_1x^2 + b_1y^2 = 16 \dots \dots (3)$$

$$a_1 + b_1 = a + b$$

$$a_1b_1 = ab - h^2$$

$$\therefore a_1 + b_1 = 5 + 5 = 10$$

$$a_1b_1 = 5.5 - (-1)^2$$

$$a_1b_1 = 24$$

$$\text{Now, } (a_1 - b_1)^2 = (a_1 + b_1)^2 - 4a_1b_1 \\ = 10^2 - 4 \cdot 24 = 100 - 96 = 4$$

$$\therefore a_1 - b_1 = 2 \text{ take +ve sign}$$

$$\therefore \begin{array}{rcl} a_1 + b_1 & = & 10 \\ a_1 - b_1 & = & 2 \end{array}$$

$$\hline (+) \quad 2a_1 = 12$$

$$\therefore a_1 = 6$$

$$b_1 = 10 - a_1 = 10 - 6 = 4$$

$$\therefore (3) \Rightarrow 6x^2 + 4y^2 = 16$$

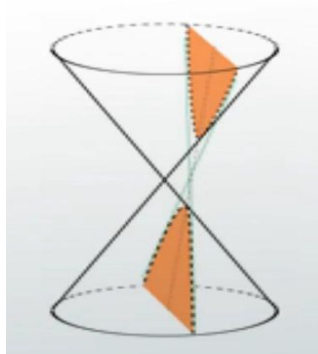
$$\Rightarrow \frac{x^2}{\frac{8}{3}} + \frac{y^2}{4} = 1$$

$$\Rightarrow \frac{x^2}{\left(\sqrt{\frac{8}{3}}\right)^2} + \frac{y^2}{y^2} = 1$$

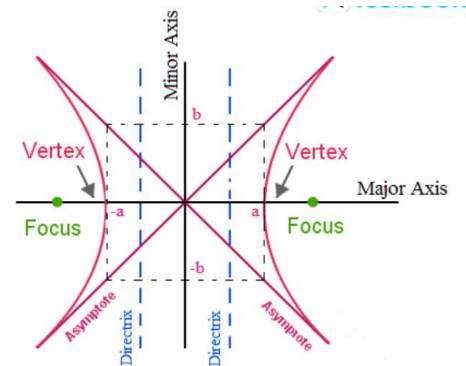
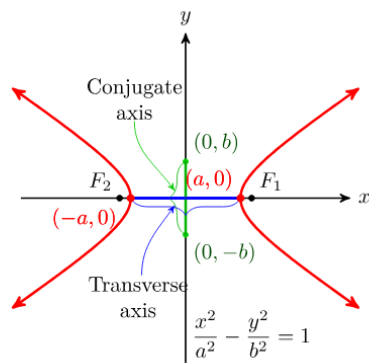
which is standard form of an ellipse.

Hyperbola

A hyperbola is a special type of conic section formed when a plane cuts through both halves of a double cone, but not exactly at the center.



A hyperbola is the set of points in the plane, the difference of whose distances from two fixed points (the foci) is a constant.



Standard equation of the hyperbola with center (h, k) and the X-axis as the transverse axis and the Y-axis as the conjugate axis is,

$$\frac{(x - h)^2}{a^2} - \frac{(y - k)^2}{b^2} = 1$$

The equation of the ellipse with center at (0,0) is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

Ex. 14(c) Reduce the equation $x^2 + 12xy - 4y^2 - 6x + 4y + 9 = 0$ to the standard form and identify it. [N.U.H. 1997]

Ans. The given equation

$$x^2 + 12xy - 4y^2 - 6x + 4y + 9 = 0 \dots (1)$$

Compare (1) with the equation

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \text{ We get}$$

$$a = 1, b = -4, h = 6, g = -3, f = 2, c = 9$$

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$$

$$= 1 \cdot (-4) \cdot 9 + 2 \cdot 2 \cdot (-3) \cdot 6 - 1 \cdot 2^2 + 4 \cdot (-3)^2 - 9 \cdot 6^2$$

$$= -36 - 72 - 4 + 36 - 324$$

$$= -400 \neq 0$$

$$ab - h^2 = 1 \cdot (-4) - 6^2 = -40 < 0$$

Thus, the given conic represents a hyperbola.

Let (α, β) be the centre of the conic (1)

$$\begin{aligned}\therefore \alpha &= \frac{hf - bg}{ab - h^2} & \beta &= \frac{hg - af}{ab - h^2} \\ &= \frac{6 \cdot 2 + 4(-3)}{1 \cdot 4 - 6^2} & &= \frac{6 \cdot 3 - 1 \cdot 2}{1 \cdot 4 - 6^2} \\ &= \frac{0}{-40} & &= \frac{-20}{-40} \\ &= 0 & &= \frac{1}{2}\end{aligned}$$

$\therefore$ Centre is at $(0, \frac{1}{2})$

Now, transfer the origine to a point $(0, \frac{1}{2})$

Then (1) becomes

$$x^2 + 12xy - 4y^2 = c_1 \dots\dots (2)$$

Where $C_1 = -(g\alpha + f\beta + C)$

$$= -(-3 \cdot 0 + 2 \cdot \frac{1}{2} + 9)$$

$$= -10$$

$$\Rightarrow x^2 + 12xy - 4y^2 = -10 \dots\dots(3)$$

Now, transform the axes such that remove the xy term, then

$$a_1 x^2 + b_1 y^2 = -10 \dots\dots(4)$$

$$\text{Where } a_1 b_1 = a + b = 1 - 4 = -3$$

$$\text{and } a_1 b_1 = ab - h^2 = -4 - 6^2 = -40$$

$$\begin{aligned}\text{Now, } (a_1 - b_1)^2 &= (a_1 + b_1)^2 - 4 a_1 b_1 \\ &= (-3)^2 - 4 \cdot -40 \\ &= 9 + 160 = 169\end{aligned}$$

$$\therefore a_1 - b_1 = 13 \quad \text{Take + ve sign}$$

$$\text{Again, } a_1 + b_1 = 3$$

$$\frac{a_1 - b_1 = 13}{(+)\ 2a_1 = 10} \quad \therefore a_1 = 5$$

$$b_1 = -3 - a_1 = -3 - 5 = -8$$

$$(4) \Rightarrow 5x^2 - 8y^2 = -10$$

$$\Rightarrow \frac{x^2}{-2} + \frac{y^2}{\frac{5}{4}} = 1$$

Which is required standard form.

Ex. 14(b) Find the asymptotes the hyperbola $x^2 + 2xy - y^2 + 2x + 4y = 0$ and the equation of the conjugate hyperbola. [N.U.H. 1996]

Ans. The given hyperbola

$$x^2 + 2xy - y^2 + 2x + 4y = 0 \dots (1)$$

Let equation of asymptote of the hyperbola (1) is

$$x^2 + 2xy - y^2 + 2x + 4y + \lambda = 0 \dots (2)$$

Since asymptotes is a pair of st. lines

$$\text{So } \Delta = 0 \text{ is } \boxed{abc + 2fgh - af^2 - bg^2 - ch^2 = 0}$$

$$\therefore 1 \cdot (-1) \cdot \lambda + 2 \cdot (+2) \cdot (+1) \cdot (-1) \cdot (+2)^2 - (-1) \cdot (+1)^2 - \lambda \cdot 1^2 = 0$$

$$\Rightarrow -\lambda + 4 - 4 + 1 - \lambda = 0$$

$$\Rightarrow 2\lambda + 1 = 0$$

$$\therefore \lambda = -\frac{1}{2}$$

$$\therefore (2) \Rightarrow x^2 + 2xy - y^2 + 2x + 4y + \frac{1}{2} = 0$$

Which is the required asymptotes.

Let equation of the conjugate hyperbola is

$$x^2 + 2xy - y^2 + 2x + 4y + k = 0 \dots (3)$$

We know, equation of the conjugate hyperbola

$$= 2 \times \{\text{equation of asymptotes}\} - \text{equation of the hyperbola}$$

$$\Rightarrow x^2 + 2xy - y^2 + 2x + 4y + k = 2 \times (x^2 + 2xy - y^2 + 2x + 4y + \frac{1}{2}) - (x^2 + 2xy - y^2 + 2x + 4y)$$

$$\Rightarrow k = 1$$

$$(3) \Rightarrow x^2 + 2xy - y^2 + 2x + 4y + 1 = 0$$

Which is the required conjugate hyperbola.

Exercise

1. Reduce the equation $x^2 + 2xy + y^2 - 2x - 1 = 0$ to its standard form and then identify it. **Ans:** Parabola; $(\frac{x+y-\frac{1}{2}}{\sqrt{2}})^2 = 4 \cdot \frac{1}{4\sqrt{2}} \cdot \frac{x-y+\frac{5}{4}}{\sqrt{2}}$;
2. Reduce the equation $8x^2 + 4xy + 5y^2 - 24x + 24y = 0$ to its standard form and then identify it. **Ans:** Ellipse; $\frac{x^2}{98/27} + \frac{y^2}{98/12} = 1$.
3. Reduce the equation $x^2 - 5xy + y^2 + 8x - 20y + 15 = 0$ to its standard form and then identify it.

Ans: Hyperbola; $\frac{x^2}{2/7} - \frac{y^2}{2/3} = 1$;